

Julia Kaczmarek

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 GitHub Profile

 LinkedIn Profile

EDUCATION

•Computational Chemistry PhD

2024-2028

The University of Manchester, Manchester, UK

Working on a project titled **Automated mechanistic studies in catalysis with quantum chemistry and machine learning** funded by the EPSRC.

•Master of Chemistry (MChem)

2021-24

Newcastle University, Newcastle-upon-Tyne, UK

First Class Honours

75.7% average

Neil Hughes Prize for contributions to the school

Levin Prize for meritorious performance in Stage 3

SCI Prize for meritorious performance in Stage 2

EXPERIENCE

•Academic Researcher in Computational Chemistry

September 2024 - Present

The University of Manchester

- Lead developer on the WebAQME and AQMEasy projects to automate computational chemistry tasks via a web app (Flask) and a desktop app (PySide6), respectively.
- Collaborated on chemical reactivity projects with two different experimental groups (organic and inorganic materials).
- Tech stack: Python (PySide6, Flask, SQLAlchemy, RDKit), HTML5, CSS, JavaScript

•Research Internship in Computational Chemistry

June 2023 - September 2023

Newcastle University

- Worked on a machine learning project which looked at using an LSTM model to predict spectroscopic lineshapes from molecular geometry data. The project was presented at two conferences (local via poster and national via talk), and the results were published in a review paper.
- Learned Linux & bash scripting, and developed a good understanding of machine learning and data science fundamentals.

PERSONAL PROJECTS

•Writing Tracker + Pomodorish Timer (Qt Framework Practice)

2024

Desktop apps to track a writing goal with approximate required words per day, and a focus timer for productivity.

- Packaged with PyInstaller. Available on GitHub (hyperlinks): Tracker and Timer.
- Technology Used : Python3, PySide6/PyQt6, PyInstaller.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, Bash, JavaScript, C

Libraries: PySide6, PyQt6, Pandas, NumPy, RDKit

Developer Tools: Git, GitHub

Systems: Linux, HPC/cluster computing

Web Dev: Flask, HTML5, CSS

Databases: SQLAlchemy

Areas of Interest: Low-level programming, app development

POSITIONS OF RESPONSIBILITY

•Theoretical Chemistry Group Committee Member

Royal Society of Chemistry

December 2024 - Current

- Appointed via election open to the entire group, which includes academics and people in industry primarily but not exclusively based in the UK .
- Organised an online scientific conference with over 150 registered attendees, 10 contributed talks, 60 contributed posters and two major plenary speakers.

•Supervisor for a Summer Student

Trujillo Research Group

June 2025 - August 2025

- Designed a project for a second year chemistry undergraduate. The project can be found on GitHub.
- Mentored and supervised the student for 6 weeks.